Nonlinear Domain Adaptation Enables Interpretable Integration of Diverse Transcriptomic Cohorts

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ABSTRACT

Integrating same-modality gene expression datasets from heterogeneous sources is essential for enhancing sample diversity, which increases statistical power for detecting different patterns in disease cohorts and allows researchers to validate findings across different populations or experimental conditions. However, this process is frequently hindered by nonlinear distributional discrepancies, or "domain shifts". Traditional linear batch-correction methods risk erasing subtle biological signals though the challenge in transcriptomic integration is not merely removing batch effects, but also preserving clinically relevant biological heterogeneity during harmonization. In this study, we propose the **Kernel Principal Component Analysis-based Domain Adaptation (KPCA-DA)**, an unsupervised framework for robust multi-source harmonization that projects genomic features into a Reproducing Kernel Hilbert Space (RKHS) to capture intricate, nonlinear gene interactions and simultaneously minimizes Maximum Mean Discrepancy (MMD) to reduce dataset-specific biases while preserving intrinsic biological heterogeneity through a maximum-variance objective. Evaluations across multiple breast cancer classification tasks demonstrate that KPCA-DA significantly improves generalization on unseen target data, outperforming both traditional batch correction and state-of-the-art deep domain adaptation methods. Importantly, KPCA-DA enables the mapping between the transformed representation and the original transcriptomic features, facilitating biologically interpretable attribution analyses of transcriptomic features. A post-hoc interpretability via Grad-CAM mapping confirms that KPCA-DA restores the prominence of canonical biomarkers such as *EGFR* and *FOXC1* on a target dataset, previously undetectable prior to adaptation. These findings, validated against independent TCGA-derived benchmarks, demonstrate that KPCA-DA preserves biological integrity while facilitating integration of scalable and interpretable biomedical data.

Keywords: Kernel-based Domain Adaptation, Transcriptomic Data Integration, Explainable AI (XAI) in Oncology, nonlinear Batch Correction, Biological Feature Restoration, Multi-source Harmonization.

In biomedical research, data integration predominantly involves the fusion of multi-modal datasets, providing complementary perspectives across diverse biological processes. However, while multi-modal integration has been extensively studied due to its ability to capture cross-domain relationships, the integration of same-modality datasets (datasets with same data type from multiple sources) remains comparatively underexplored. Same-modality integration is particularly crucial for merging datasets from different studies to enlarge sample sizes, thereby increasing statistical power, facilitating the discovery of subtle molecular patterns, and strengthening reproducibility through validation across independently collected cohorts. For example, the integration of RNA sequencing (RNA-seq) datasets from multiple studies investigating similar biological conditions[1] or the aggregation of single-cell RNA-sequencing data from large-scale tissue atlases[2] provides a more comprehensive view of gene expression landscapes. However, a major challenge in multi-source same-modality data integration is the distributional heterogeneity between datasets that arises due to variations in experimental protocols, sequencing platforms, sample processing techniques, and batch effects[3,4]. These discrepancies hinder direct dataset fusion, as models trained on one dataset may generalize poorly to another with distinct distributional characteristics. For instance, RNA-seq datasets generated in different laboratories can exhibit systematic shifts in gene expression profiles, complicating direct comparisons and integration. Addressing these variations requires mathematical and statistical approaches to quantify and minimize distributional differences while preserving biologically meaningful signals.

Existing domain discrepancy reduction methods for genomic data like batch correction (e.g., ComBat[5], Limma[6]) and multi-view integration ([7–9]) are often linearity-based, making them less effective for complex, nonlinear genomic datasets such as RNA-seq. Since gene interactions and biological pathways are not yet fully understood, assuming linearity in microarray sample spaces and gene connectivity is often misleading[10,11]. A more effective method to alleviate such discrepancies is domain adaptation (DA), which enables models trained on a source dataset to generalize effectively to a target dataset despite inherent distributional differences. Unlike conventional batch correction methods, domain adaptation approaches explicitly quantify the distributional shift and employ transformation strategies to minimize the shifts. DA tech-

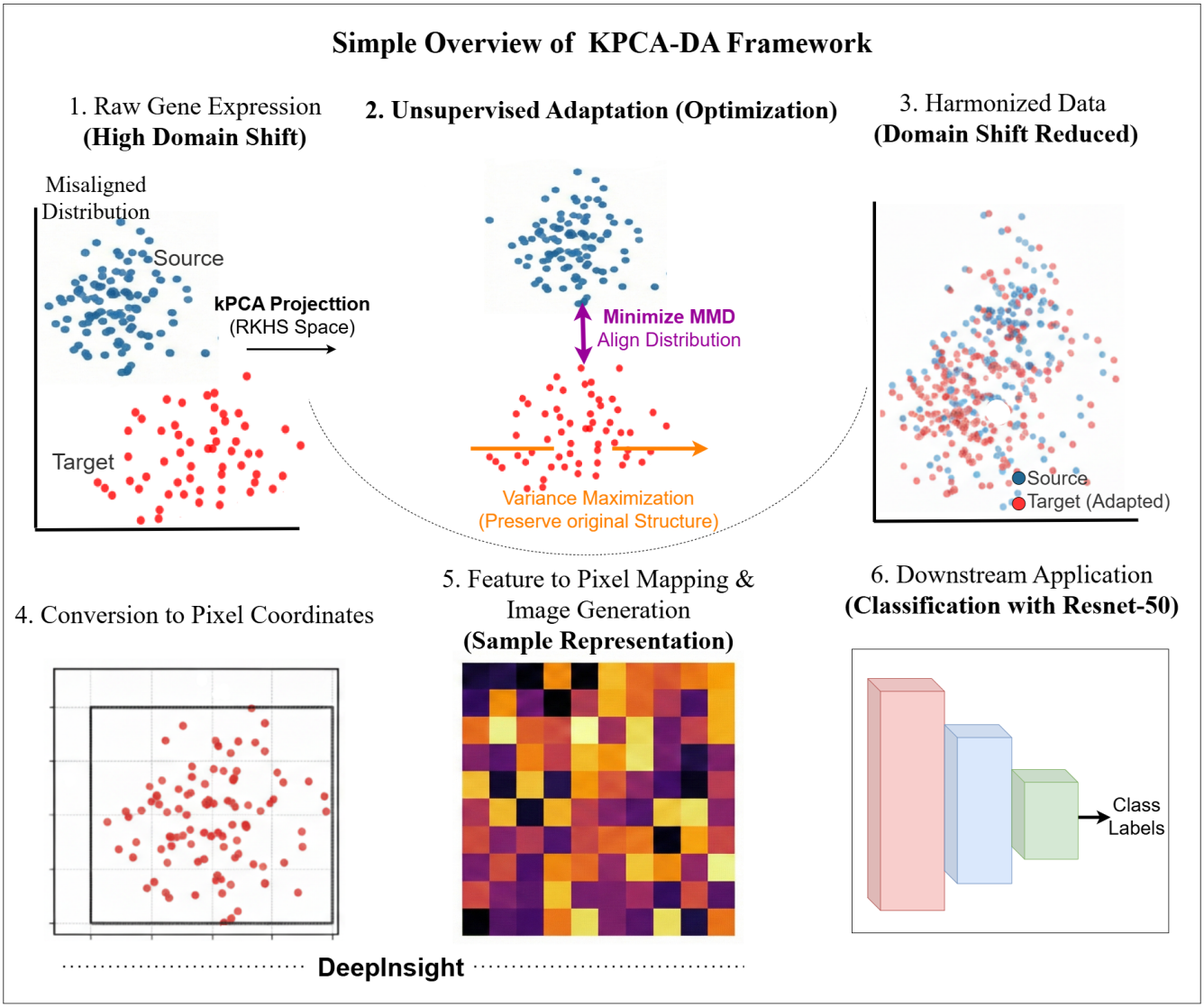


Figure 1. Schematic overview of the KPCA-DA framework for nonlinear gene expression integration. The multi-staged pipeline addresses distributional heterogeneity in same-modality transcriptomic datasets through unsupervised feature alignment. **(1)** Raw source and target gene expression data exhibit significant domain shifts arising from disparate experimental protocols or platforms. **(2, 3)** Data are projected into a Reproducing Kernel Hilbert Space (RKHS) via **kPCA**, where unsupervised domain adaptation simultaneously minimizes distributional discrepancies—quantified by **Maximum Mean Discrepancy (MMD)**—and preserves discriminative biological structures through **variance maximization**. **(4, 5)** Harmonized features are converted into fixed pixel coordinates and transformed into 2D image-like representations using the **DeepInsight** algorithm, enabling structured sample encoding. **(6)** These synthetic images are utilized for downstream predictive tasks, such as molecular subtyping or prognostic classification, using a **ResNet-50** convolutional neural network architecture.

niques are particularly advantageous for nonlinear biological data, as they leverage advanced mathematical transformations to align feature spaces across datasets. Similar advances in nonlinear representation learning have been demonstrated in genomic foundation models, where biologically meaningful sequence representations are learned that generalize across diverse datasets, highlighting the importance of rich latent embeddings for cross-dataset transferability[12]. DA methods apply various mathematical measurements such as the maximum mean discrepancy (MMD)[13] to quantify the distribution difference in datasets. However, to apply MMD to gene expression datasets such as mRNA, the data first need to be mapped to an appropriate kernel space where nonlinear relationships between genes can be meaningfully captured. Kernel-based DA strategies, particularly those leveraging Maximum Mean Discrepancy (MMD)[13], provide a principled way to quantify and reduce domain shifts while enabling nonlinear transformations suited to genomic data. While kPCA projects

high-dimensional gene expression data into a low-dimensional latent space that captures complex, nonlinear biological relationships, MMD serves as a statistical constraint to minimize distributional discrepancies across datasets within this projected space[3]. Such a combination ensures that latent representations not only preserve biologically meaningful variance but also align source and target datasets, thereby reducing domain shifts introduced by heterogeneous experimental protocols or patient cohorts[14].

In this study, we propose KPCA-DA (Kernel Principal Component Analysis-Guided Domain Adaptation), an unsupervised domain adaptation framework designed to integrate heterogeneous gene expression datasets. Kernel Principal Component Analysis (kPCA) extends traditional PCA by projecting data into a high-dimensional feature space using kernel functions, which can capture complex, nonlinear relationships within genomic data. While originally developed for dimensionality reduction, kPCA is highly effective in domain adaptation by mapping datasets into a shared kernel space, facilitating alignment between different experimental conditions, sequencing platforms, or batches while preserving underlying biological signals. Kernel PCA (kPCA) employs kernel functions to implicitly project data into a high-dimensional feature space, followed by dimensionality reduction into a low-dimensional latent space that captures nonlinear structures often missed by linear methods. This enables the preservation of biologically meaningful patterns and supports downstream tasks such as clustering, visualization, and classification. The adaptation process with KPCA-DA is constrained by incorporating Maximum Mean Discrepancy (MMD), a statistical similarity measurement method that quantifies the distributional distance between datasets in a Reproducing Kernel Hilbert Space (RKHS)[3]. This constraint encourages the projected source and target data to share a similar distribution in the latent space. By minimizing MMD in the projected feature space, our approach systematically reduces domain shift while maintaining biologically relevant variations. Moreover, we introduce a maximum variance loss function to retain domain-specific information, ensuring that meaningful biological information in target dataset is not lost in the alignment process (fig-1). Our approach is inspired by DeepInsight[15], a method that converts high-dimensional omics data into image-like representations for deep learning analysis. We extend this concept by leveraging kPCA for domain alignment, transforming gene expression data into structured 2D feature representations, which are subsequently classified using ResNet-50, a deep convolutional neural network (fig-1). By evaluating classification performance across datasets, we assess whether KPCA-DA effectively harmonizes genomic datasets, enabling cross-study comparisons and improving predictive accuracy on unseen data. Through this study, we aim to demonstrate that KPCA-DA provides a robust, fully unsupervised framework for harmonizing genomic datasets, particularly for deep learning applications in bioinformatics. By addressing nonlinear domain shifts, our approach enhances dataset interoperability, paving the way for more reproducible and scalable biomedical analysis.

Results

Experiments were conducted to evaluate the effectiveness of the proposed KPCA-DA framework in integrating and classifying breast cancer mRNA datasets across multiple sources. The experimental settings were designed based on the three classification tasks using multiple publicly available breast cancer mRNA expression datasets. The datasets were retrieved from The Cancer Genome Atlas (TCGA), Gene Expression Omnibus (GEO), CBioPortal, Clinical Proteomic Tumor Analysis Consortium (CPTAC), and the study by Desmedt et al. (2007)[16–18]. Gene expression values were standardized to ensure compatability across datasets.

Dataset Description

Task 1: Aggressive vs. Indolent Breast Cancer Classification : The first classification objective distinguishes aggressive from indolent breast cancer phenotypes based on transcriptomic profiles. The datasets for this task were integrated from the TCGA BRCA and *Desmedt et al.* (2007) cohorts[16]. Clinical labels were established using disease-specific survival (DSS) thresholds: patients with survival exceeding 10 years were categorized as indolent, while those who succumbed to the disease within five years were classified as aggressive. This stratified approach adheres to the prognostic criteria established by *Desmedt et al.*, where five-year survival serves as a proxy for poor prognosis and ten-year survival indicates favorable outcomes. The TCGA cohort comprises 160 patients (18,000 gene features), while the Desmedt dataset includes 164 patients (9,000 gene features). To ensure cross-platform consistency, genes with zero expression across all samples were removed prior to harmonization.

Task 2: PAM50 Molecular Subtypes Classification : The second classification objective focuses on molecular subtyping using the PAM50 system, also known as the Prosigna Breast Cancer Prognostic Gene Signature Assay. This assay stratifies breast cancer into four intrinsic molecular subtypes: Luminal A, Luminal B, HER2-enriched, and Basal-like[19]. For this evaluation, two mRNA expression datasets with corresponding PAM50 annotations were retrieved from CBioPortal. The source domain originates from the Korean Breast Cancer cohort (SMC) ($n = 166$)[17], while the target domain was sourced from the Clinical Proteomic Tumor Analysis Consortium (CPTAC) ($n = 117$)[18]. These cohorts provide a diverse representation of molecular landscapes, facilitating a robust assessment of the proposed framework's capacity for cross-domain subtyping. While a small subset of Normal-like samples was present in the raw data, they were excluded from the analysis due to insufficient sample size for meaningful statistical inference.

Task 3: Tumor vs. Normal Tissue Classification : The third classification objective involves the binary differentiation of malignant tumor tissue from tumor-adjacent normal breast tissue. Utilizing tumor-adjacent normal samples as a control group is standard practice for identifying the distinct

Table 1. Comparative performance of domain adaptation methods across diagnostic and prognostic breast cancer classification tasks. Summary of classification metrics (Accuracy, F1-score, Precision, Sensitivity, Specificity, and Balanced Accuracy) for three distinct tasks: (i) *Aggressive vs. Indolent Survival*, (ii) *PAM50 Molecular Subtyping*, and (iii) *Tumor vs. Normal Tissue Differentiation*. The proposed **KPCA-DA** framework is compared against a non-adapted baseline (**W/O DA**), statistical batch correction (**ComBat**), and deep learning-based unsupervised domain adaptation (**CAN**). Models were trained on a source domain and evaluated on a held-out 30% subset of the target domain to rigorously test cross-study generalization. **Green shading** indicates the top-performing model for each metric, while **yellow shading** denotes the second-best performance. The results demonstrate that **KPCA-DA** consistently achieves state-of-the-art performance, outperforming linear and image-domain adaptation methods across all clinical benchmarks.

Task	Train Data	Test Data	Model	Acc.	F1	Prec.	Sens.	Spec.	Bal. Acc.
Aggressive vs. Indolent Disease Classification	TCGA	Desmedt	W/O DA	0.58 ± 0.04	0.63 ± 0.04	0.60 ± 0.03	0.64 ± 0.03	0.55 ± 0.04	0.60 ± 0.02
			ComBat	0.65 ± 0.02	0.68 ± 0.03	0.64 ± 0.01	0.76 ± 0.02	0.60 ± 0.02	0.66 ± 0.02
			CAN	0.68 ± 0.02	0.67 ± 0.03	0.68 ± 0.02	0.72 ± 0.02	0.67 ± 0.02	0.68 ± 0.01
			KPCA-DA	$.70 \pm 0.02$	0.72 ± 0.02	0.71 ± 0.01	0.74 ± 0.02	0.68 ± 0.02	0.71 ± 0.01
	Desmedt	TCGA	W/O DA	0.56 ± 0.05	0.60 ± 0.06	0.70 ± 0.05	0.57 ± 0.05	0.68 ± 0.04	0.60 ± 0.05
			ComBat	0.61 ± 0.04	0.64 ± 0.05	0.76 ± 0.04	0.60 ± 0.03	0.71 ± 0.04	0.64 ± 0.02
			CAN	0.63 ± 0.02	0.67 ± 0.03	0.77 ± 0.04	0.62 ± 0.02	0.72 ± 0.03	0.65 ± 0.03
			KPCA-DA	0.66 ± 0.02	0.71 ± 0.03	0.80 ± 0.02	0.64 ± 0.01	0.74 ± 0.02	0.69 ± 0.02
PAM50 Subtypes Classification	SMC	CPTAC	W/O DA	0.63 ± 0.02	0.61 ± 0.02	0.65 ± 0.03	0.65 ± 0.02	0.62 ± 0.01	0.64 ± 0.02
			ComBat	0.66 ± 0.03	0.66 ± 0.03	0.67 ± 0.02	0.70 ± 0.04	0.66 ± 0.04	0.68 ± 0.02
			CAN	0.68 ± 0.02	0.69 ± 0.03	0.65 ± 0.04	0.74 ± 0.05	0.65 ± 0.06	0.70 ± 0.04
			KPCA-DA	0.70 ± 0.03	0.71 ± 0.04	0.70 ± 0.05	0.72 ± 0.04	0.68 ± 0.03	0.73 ± 0.03
	CPTAC	SMC	W/O DA	0.61 ± 0.03	0.62 ± 0.03	0.65 ± 0.04	0.66 ± 0.04	0.60 ± 0.03	0.62 ± 0.02
			ComBat	0.65 ± 0.02	0.67 ± 0.02	0.66 ± 0.03	0.70 ± 0.02	0.62 ± 0.04	0.67 ± 0.02
			CAN	0.67 ± 0.02	0.70 ± 0.05	0.68 ± 0.05	0.72 ± 0.06	0.66 ± 0.05	0.68 ± 0.03
			KPCA-DA	0.68 ± 0.03	0.70 ± 0.02	0.71 ± 0.02	0.69 ± 0.04	0.73 ± 0.04	0.70 ± 0.02
Tumor vs. Normal Tissue Classification	TCGA	GEO	W/O DA	0.67 ± 0.04	0.60 ± 0.03	0.70 ± 0.03	0.80 ± 0.04	0.62 ± 0.05	0.64 ± 0.03
			ComBat	0.70 ± 0.03	0.76 ± 0.03	0.75 ± 0.04	0.84 ± 0.03	0.60 ± 0.02	0.69 ± 0.02
			CAN	0.75 ± 0.02	0.78 ± 0.03	0.78 ± 0.02	0.85 ± 0.02	0.64 ± 0.02	0.72 ± 0.03
			KPCA-DA	0.77 ± 0.03	0.82 ± 0.04	0.80 ± 0.03	0.85 ± 0.03	0.65 ± 0.02	0.75 ± 0.03
	GEO	TCGA	W/O DA	0.65 ± 0.02	0.73 ± 0.04	0.70 ± 0.06	0.68 ± 0.05	0.62 ± 0.05	0.63 ± 0.03
			ComBat	0.70 ± 0.03	0.74 ± 0.02	0.72 ± 0.04	0.77 ± 0.03	0.64 ± 0.02	0.69 ± 0.02
			CAN	0.73 ± 0.01	0.76 ± 0.02	0.73 ± 0.03	0.81 ± 0.03	0.66 ± 0.02	0.70 ± 0.02
			KPCA-DA	0.75 ± 0.02	0.78 ± 0.05	0.76 ± 0.04	0.79 ± 0.03	0.70 ± 0.04	0.74 ± 0.04

genomic signatures associated with oncogenesis. This task serves as a critical baseline to evaluate the effectiveness of the KPCA-DA framework in capturing high-variance biological differences across disparate data sources. The cohorts for this analysis were integrated from TCGA ($n = 592$) and a curated GEO (GSE45498) dataset ($n = 224$). Given that tumor and normal tissues typically exhibit fundamental transcriptomic differences, this task provides a benchmark for assessing whether the domain adaptation process preserves these biological boundaries while mitigating platform-induced technical variations.

Quantitative Assessment of Domain Alignment

The efficacy of domain adaptation in genomics is measured by its ability to align target and source distributions while mitigating study-specific artifacts that confound downstream classification. To benchmark **KPCA-DA**, we selected two representative yet distinct paradigms: **ComBat**[5], the gold-standard statistical batch correction method, and the **Contrastive Adaptation Network (CAN)**[20], a state-of-the-art unsupervised deep learning framework. ComBat served as the statistical baseline, however, its requirement for identical feature sets necessitated the removal of non-intersecting genes across datasets prior to alignment. In contrast, CAN—originally optimized for computer vision—was selected to represent the deep-learning-based domain adaptation paradigm. By comparing KPCA-DA against both a linear statistical method (ComBat) and a contrastive deep learning architecture (CAN), we rigorously evaluated our framework's capacity to capture nonlinear biological signals that traditional methods might obscure.

The classification performance across all three clinical benchmarks is summarized in Table 1, with the top performance indicated in green and second-best in yellow. Across all evaluated scenarios, the integration of domain adaptation led to significant performance gains over the non-adapted baseline ($W/O-DA$). Notably, **KPCA-DA** consistently emerged as the top-performing framework, particularly in the most challenging task: predicting *Aggressive vs. Indolent Survival*. In the

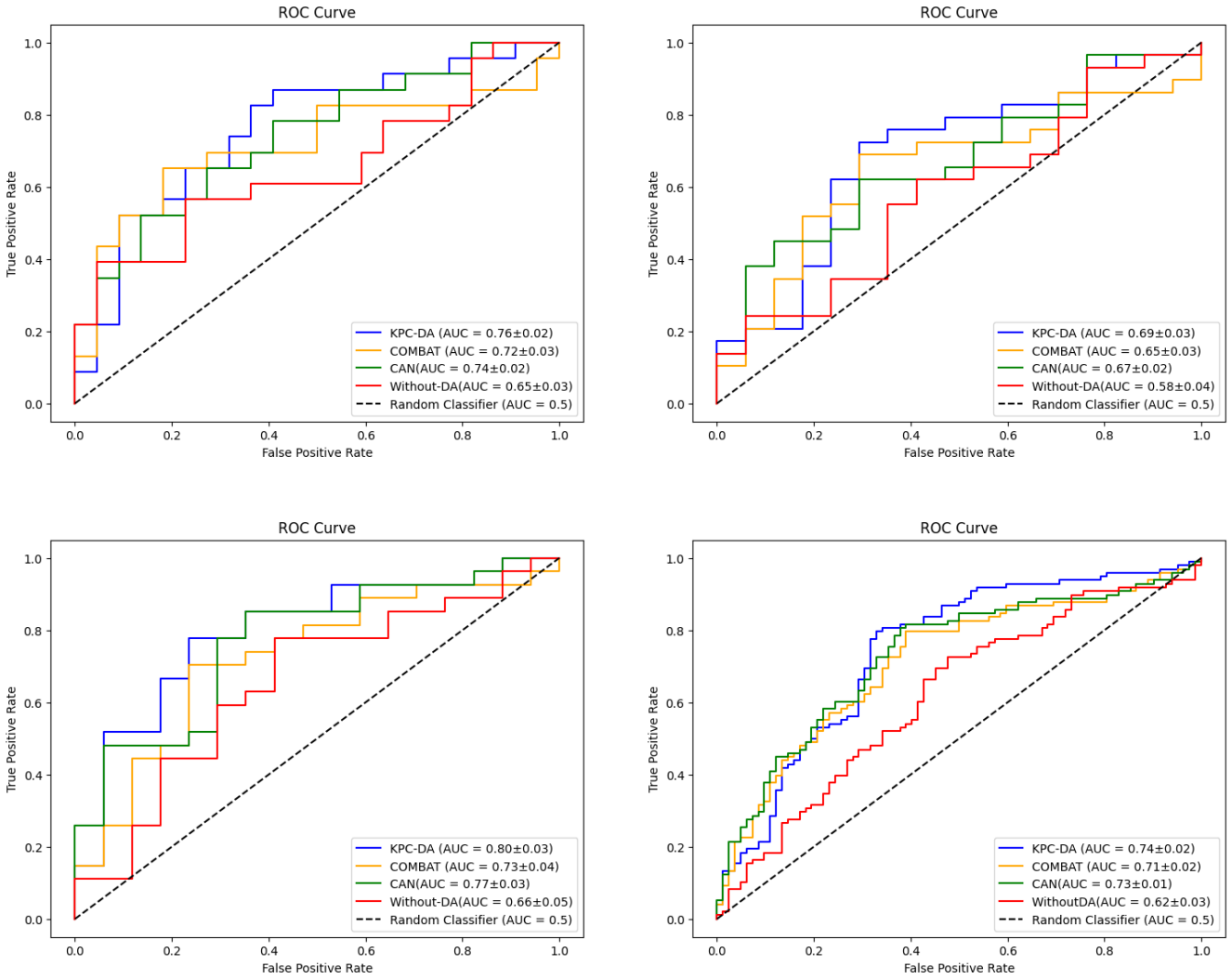


Figure 2. Robustness of KPCA-DA across bi-directional cross-domain diagnostic and prognostic tasks. Receiver Operating Characteristic (ROC) analysis evaluating the framework's performance under reciprocal domain shifts compared to state-of-the-art methods. **(a, b)** Prognostic classification (Aggressive vs. Indolent) conducted for both the *TCGA* → *Desmedt* shift and the reverse *Desmedt* → *TCGA* shift. **(c, d)** Diagnostic differentiation (Tumor vs. Normal) for the *TCGA* → *GEO* and *GEO* → *TCGA* cohorts. In all reciprocal tests, KPCA-DA (navy blue) maintains superior AUC stability compared to linear batch correction (ComBat), deep-learning based adaptation (CAN), and the non-adapted baseline. These results demonstrate that kernel-based alignment effectively captures nonlinear domain shifts, ensuring robust model generalization regardless of which independent study serves as the source.

TCGA → *Desmedt* setup, KPCA-DA achieved an accuracy of 70%, surpassing ComBat and CAN by 5% and 2%, respectively. This superior generalization was maintained in the reverse experiment (*Desmedt* → *TCGA*), where KPCA-DA (66%) again outperformed both deep learning and statistical baselines.

For the four-way *PAM50 molecular subtyping* task, KPCA-DA demonstrated high sensitivity to subtle transcriptomic distinctions, achieving an average accuracy of 70% (*SMC* → *CPTAC*), a marked improvement over CAN and ComBat. The framework's robustness was further confirmed in the *Tumor vs. Normal* baseline task. When adapting from *TCGA* → *GEO*, KPCA-DA reached a peak accuracy of 77%, a 10% absolute improvement over the non-adapted baseline. These results, corroborated by the ROC curves and AUC scores (Figure 2), demonstrate that KPCA-DA provides a more reliable and comprehensive harmonization across heterogeneous genomic datasets than existing state-of-the-art methods. To confirm that observed performance gains were not due to chance, we conducted pairwise Wilcoxon signed-rank tests between KPCA-

DA and each baseline method across all tasks and metrics where all reported improvements were statistically significant (p-values ranging from 0.03 to 0.035).

Ablation Analysis of Framework Components

The quantitative evaluation of the ablation configurations across all three diagnostic and prognostic benchmarking tasks demonstrates that every architectural component within the KPCA-DA framework plays an indispensable role in cross-study harmonization. As summarized in Table 2, eliminating the active domain alignment completely (*kPCA + VML*) forces a significant drop in downstream accuracy, collapsing performance toward the unadapted baseline due to unmitigated platform discrepancies. Conversely, executing domain alignment without structural constraints (*kPCA + MMD*) also degrades performance across all configurations. Most notably, in the highly volatile prognostic task (TCGA → Desmedt), omitting the Variance Maximization Loss (VML) decreases accuracy from $\mathbf{0.70 \pm 0.02}$ down to 0.64 ± 0.02. This drop directly exposes the vulnerability of distance-minimization metrics to localized mode collapse when unshared feature structures are not adequately preserved. Strikingly, utilizing feature mapping in isolation (*kPCA Only*) yields the lowest generalizability across all cohorts, confirming that latent projection alone is insufficient without explicit distribution optimization. Maximum performance is only unlocked when nonlinear kernel transformations, statistical alignment, and variance regularization are synergistically executed by the full framework.

Table 2. Ablation study evaluating the impact of core KPCA-DA architectural components across various studies. Best results are indicated in bold.

Model Variant	TCGA → GEO	TCGA → Desmedt	CPTAC → SMC
kPCA Only (No MMD, No VML)	0.67 ± 0.04	0.58 ± 0.04	0.61 ± 0.04
kPCA + MMD (No VML Regularization)	0.72 ± 0.04	0.64 ± 0.02	0.65 ± 0.02
kPCA + VML (No MMD Alignment)	0.69 ± 0.02	0.61 ± 0.05	0.62 ± 0.03
Full KPCA-DA	$\mathbf{0.77 \pm 0.03}$	$\mathbf{0.70 \pm 0.02}$	$\mathbf{0.68 \pm 0.03}$

Evaluation of Distributional Shift Reduction

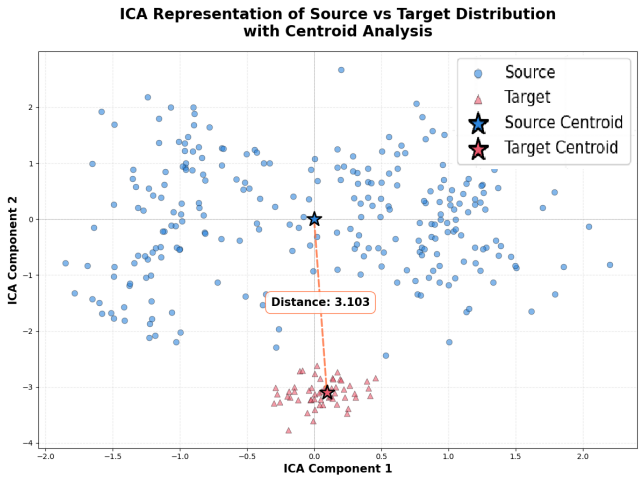
In addition to classification performance, we assessed the alignment between the source and target domains using Independent Component Analysis (ICA) for both qualitative visualization and quantitative evaluation. ICA was employed as a linear projection technique to capture statistically independent components within the genomic data. Specifically, an ICA model was fitted to the source dataset, and the learned projection was subsequently applied to the target-test set, embedding both domains into a shared coordinate space. This approach ensured a consistent representation, enabling a direct comparison of the degree of overlap between distributions before and after adaptation. Geometric alignment was quantified by computing the Euclidean distance between the centroids of the source and target embeddings and a reduction in this distance serves as a primary indicator of successful cluster harmonization. To complement this geometric analysis, we utilized the *Wasserstein Distance* (WD) to provide a robust quantitative measure of distributional similarity. The results of the ICA-based manifold alignment and subsequent WD calculations utilizing the TCGA (source) and GEO (target) datasets from the Tumor vs. Normal (Task 3) classification as a representative benchmark for cross-domain harmonization, are presented in Table 3 and Figure 3. The Tumor vs. Normal task (Task 3) was selected as the primary representative benchmark for the ICA visualization in Figure 3 for two methodologically grounded reasons. First, this task involves the largest combined cohort (TCGA n=592, GEO n=224), providing the most statistically stable embedding for centroid distance estimation. Second, as a binary classification task with well-established biological boundaries between tumor and normal tissue, it provides the clearest ground truth against which the quality of distributional alignment can be visually and quantitatively interpreted. The baseline (W/O DA) revealed a substantial discrepancy ($\text{WD} = 2.086 \pm 0.561$), underscoring the severity of study-specific artifacts in unharmonized data. While all tested adaptation frameworks reduced this gap, **KPCA-DA** achieved the most profound alignment, yielding a WD of $\mathbf{0.584 \pm 0.312}$. This represents a **72.0% reduction** in distributional discrepancy relative to the baseline, significantly outperforming CAN (58.0% reduction) and ComBat (50.3% reduction). These gains are visually corroborated by the projections in Fig. 3, where KPCA-DA effectively collapses inter-study variance while preserving intra-class biological structure. These findings suggest that the integration of nonlinear kernel mapping and Maximum-Variance Loss is uniquely capable of resolving the complex, high-dimensional shifts inherent in transcriptomic datasets.

Table 3. Quantitative assessment of domain alignment via Wasserstein distance reduction. Distributional discrepancy between source and target domains is quantified using the Wasserstein distance (WD) in a shared 2D embedding space. Percentage reduction (Δ) is calculated relative to the unadapted Baseline. Best results are indicated in **bold**.

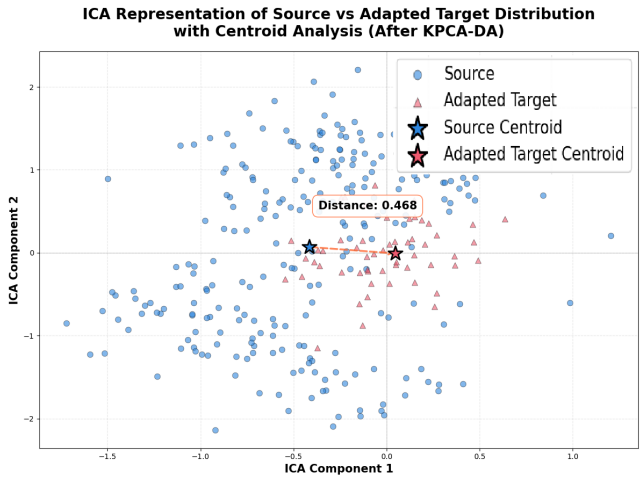
Method	WD ($\mu \pm \sigma$)	Δ (%)
Baseline	2.086 ± 0.561	—
ComBat[5]	1.037 ± 0.429	50.3
CAN[20]	0.875 ± 0.348	58.0
KPCA-DA	$\mathbf{0.584 \pm 0.312}$	**72.0**

White-Box Interpretability and Validation of Feature Recovery

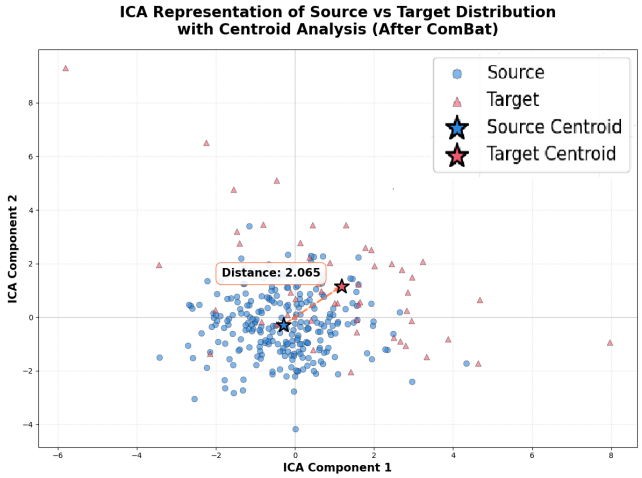
To resolve the "black-box" challenge inherent in kernel-based methods, we utilized Grad-CAM[21] to map model attributions



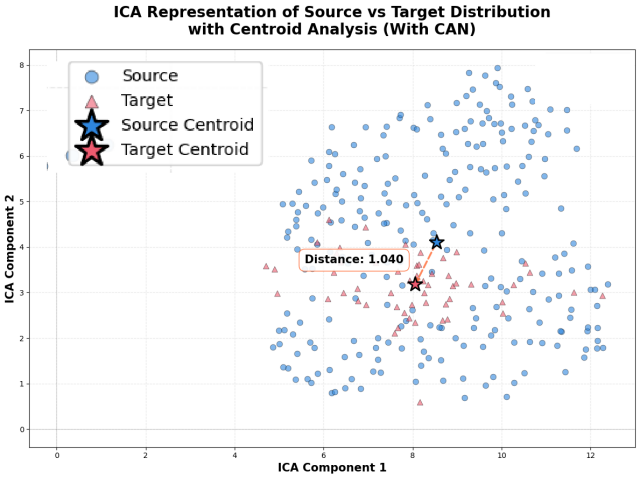
(a) WD=2.0371



(b) WD=.7010



(c) WD=1.4372



(d) WD=1.1139

Figure 3. Manifold alignment and quantitative reduction of distributional discrepancy. Comparative analysis of domain harmonization using Independent Component Analysis (ICA) projections and centroid distance metrics (Source: TCGA, Target: GEO). **(a) Baseline (Before DA):** Displays a substantial initial domain shift ($WD = 2.0371$) with a large separation (3.103) between the source (blue) and target (red) centroids. **(b) After KPCA-DA (Proposed):** Demonstrates optimal manifold alignment, achieving the lowest centroid distance (0.468) and a significant reduction in distributional discrepancy ($WD = 0.7010$), representing an approximate 72% improvement over the baseline. **(c) After ComBat:** Shows the limitations of linear batch correction, which leaves a larger residual gap ($WD = 1.4372$). **(d) With CAN:** Illustrates a deep-learning based alignment that, while superior to the baseline, is less effective at capturing nonlinear biological manifolds than the proposed kernel-based method ($WD = 1.1139$)

from the image-domain back to the original transcriptomic space. This enables a "white-box" transition, allowing us to determine if **KPCA-DA** harmonizes datasets by preserving real biological drivers rather than study-specific noise. For each PAM50 subtype, Grad-CAM importance maps were computed across the target samples of that subtype. Each pixel was mapped to its corresponding gene using the fixed DeepInsight coordinate assignment, and genes were ranked by their mean spatial Grad-CAM activation score. The top-10 genes per subtype (n=40 total) were retained for downstream overlap analysis. This gene-to-pixel correspondence has been leveraged in studies[15,22] to enable feature-level interpretability through CNN-based gradient methods. We conducted this interpretability analysis on the PAM50 sub-typing task (SMC $\rightarrow$ CPTAC), systematically comparing the performance of **KPCA-DA**, **ComBat**, and the unadapted **baseline**.

To objectively "grade" the model's feature discovery, we established two independent benchmarks. The first is the canonical **PAM50 signature** (50 genes). The second is a "ground truth" set of 18 genes identified via differential expression (DE) analysis of an independent ***TCGA-BRCA*** cohort using *iPathwayGuide* (Advaita Bioinformatics). These 18 genes exhibit

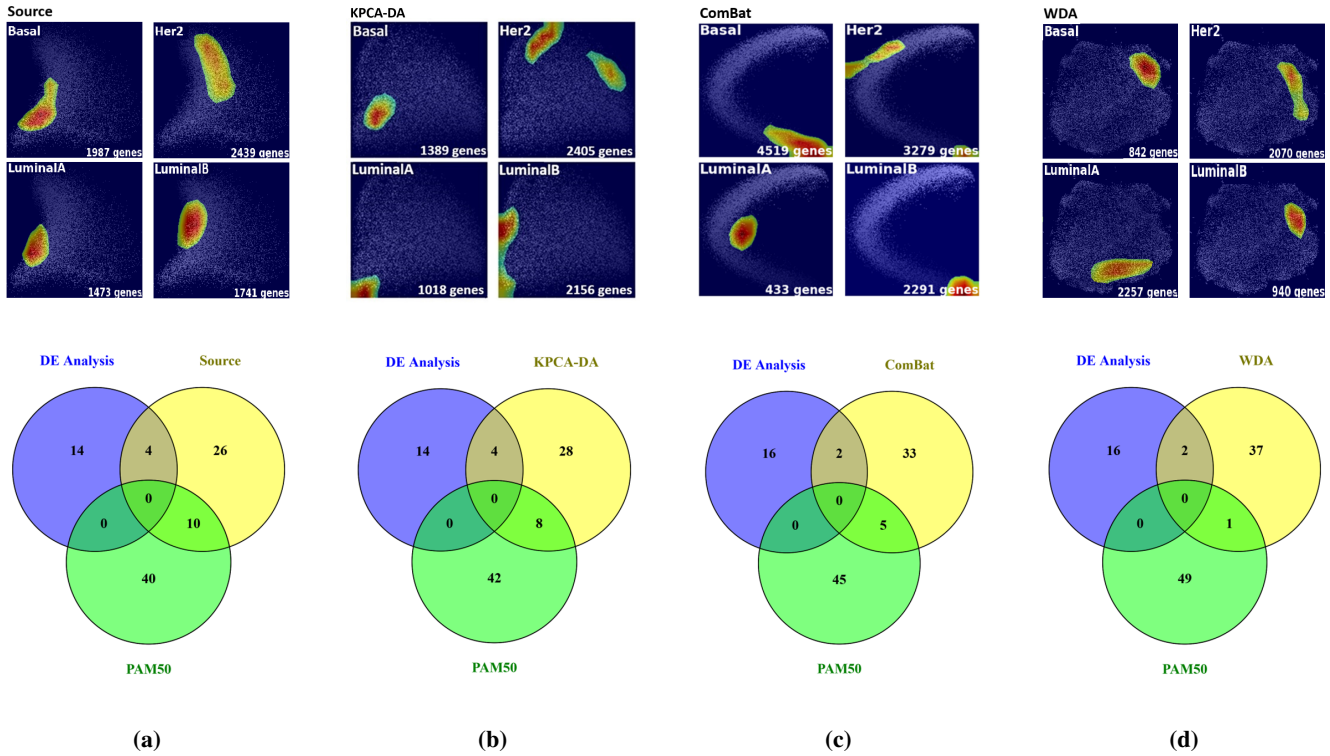


Figure 4. Biological validation through feature attribution and canonical biomarker recovery. Assessment of biological signal preservation using Grad-CAM heatmaps and multi-source gene overlap analysis. **(a) Source Reference:** Establishes the primary biological drivers identified in the original SMC training cohort. **(b–d) Target Adaptation:** Comparison of feature attribution on the CPTAC target cohort using **(b) KPCA-DA**, **(c) ComBat**, and **(d) Baseline (no adaptation)**. Grad-CAM heatmaps visualize the gene clusters most influential to the model's subtype predictions. Corresponding Venn diagrams quantify the overlap between model-selected features, differentially expressed (DE) genes from *iPathwayGuide*, and the clinically validated **PAM50 gene signature**. Notably, **KPCA-DA (b)** recovers a significantly higher proportion of ground-truth biomarkers compared to traditional methods. This demonstrates the framework's *"white-box" interpretability*, enabling the direct mapping of aligned data representations back to the specific biological drivers of disease.

exclusive up- or down-regulation within specific subtypes and are detailed in **Figure 5**, which illustrates their presence across the SMC (source) and CPTAC (target) datasets.

From the model's output, we extracted the **top 10 most influential genes per subtype** ($n = 40$) based on Grad-CAM importance scores. We then quantified the overlap between these model-identified features and our two benchmarks. As shown in **Figure 4**, **KPCA-DA** yielded attributions that were visually and quantitatively more consistent with the source domain than the alternatives. Notably, KPCA-DA recovered a significantly higher proportion of both the PAM50 signature and the TCGA-derived DE genes, confirming that our domain adaptation process successfully restores subtype-specific markers that were previously obscured by domain shift.

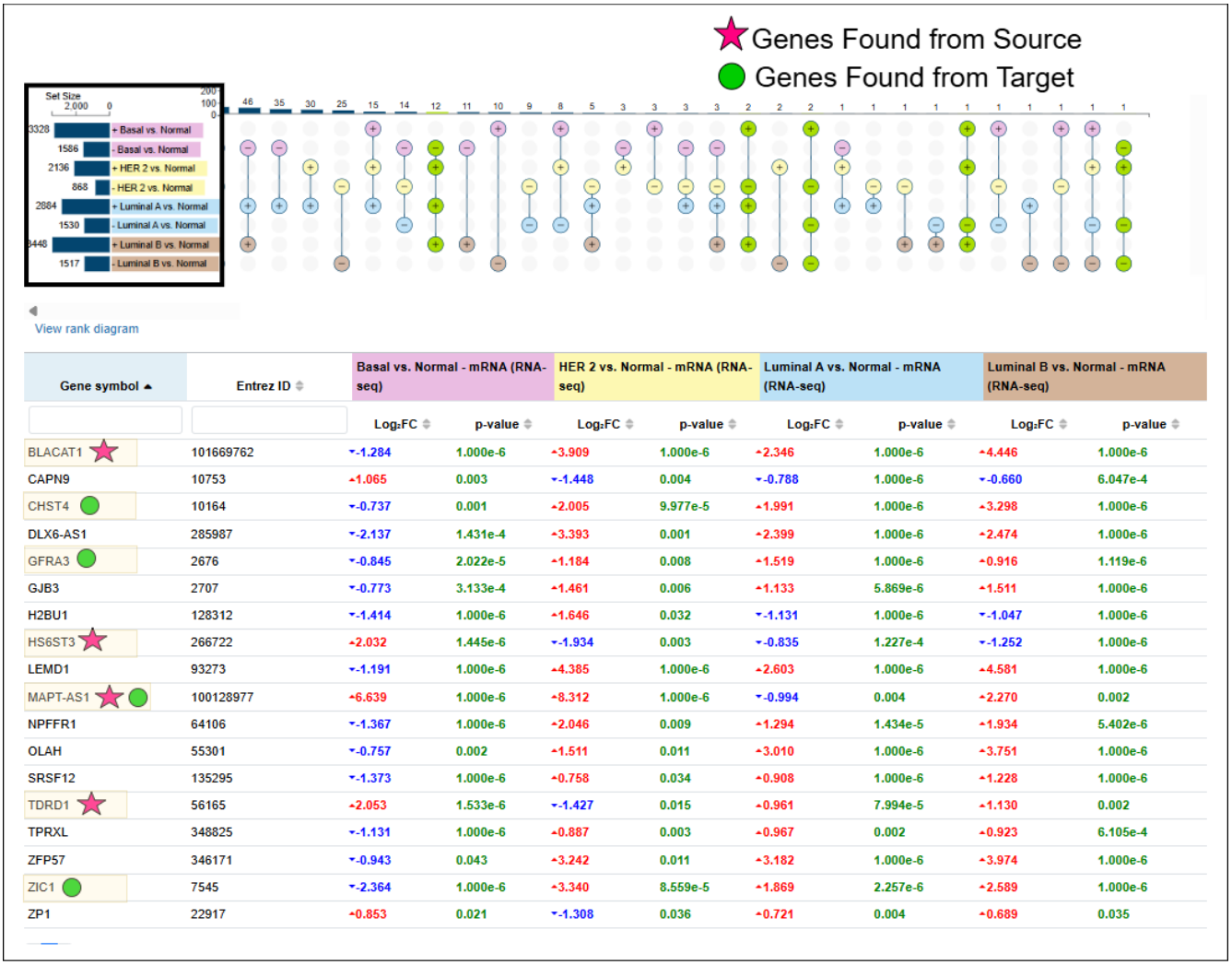
We excluded the **Contrastive Adaptation Network (CAN)** from this specific interpretability analysis. Because CAN operates in a highly abstract latent space optimized solely for domain-invariance, the explicit mapping between individual input dimensions (genes) and learned features is discarded. In contrast, the spatial preservation of our framework allows for a "white-box" transition, directly linking harmonized representations back to the specific biological drivers of disease (Table 4).

Discussion

The integration of high-dimensional transcriptomics into clinical decision-making has historically been hindered by what we term the *Batch Effect Paradox*: aggressive statistical correction often inadvertently erases the subtle biological signals required for precision medicine. Our results demonstrate that **KPCA-DA** resolves this paradox by shifting the harmonization paradigm from linear distribution matching to nonlinear manifold alignment.

The Nonlinear Advantage and Variance Preservation

Traditional batch correction methods, exemplified by ComBat[5], operate on the assumption that inter-study variation is a linear additive or multiplicative bias. However, our observation of a 72.0% reduction in Wasserstein Distance—significantly higher than the alignment achieved by ComBat (50.3%) or



Gene symbol	Entrez ID	Basal vs. Normal - mRNA (RNA-seq)		HER 2 vs. Normal - mRNA (RNA-seq)		Luminal A vs. Normal - mRNA (RNA-seq)		Luminal B vs. Normal - mRNA (RNA-seq)	
		Log_2FC	p-value	Log_2FC	p-value	Log_2FC	p-value	Log_2FC	p-value
BLACAT1 ★	101669762	-1.284	1.000e-6	3.909	1.000e-6	2.346	1.000e-6	4.446	1.000e-6
CAPN9	10753	1.065	0.003	-1.448	0.004	-0.788	1.000e-6	-0.660	6.047e-4
CHST4 ●	10164	-0.737	0.001	2.005	9.977e-5	1.991	1.000e-6	3.298	1.000e-6
DLX6-AS1	285987	-2.137	1.431e-4	3.393	0.001	2.399	1.000e-6	2.474	1.000e-6
GFRA3 ●	2676	-0.845	2.022e-5	1.184	0.008	1.519	1.000e-6	0.916	1.119e-6
GJB3	2707	-0.773	3.133e-4	1.461	0.006	1.133	5.869e-6	1.511	1.000e-6
H2BU1	128312	-1.414	1.000e-6	1.646	0.032	-1.131	1.000e-6	-1.047	1.000e-6
HS6ST3 ★	266722	2.032	1.445e-6	-1.934	0.003	-0.835	1.227e-4	-1.252	1.000e-6
LEMD1	93273	-1.191	1.000e-6	4.385	1.000e-6	2.603	1.000e-6	4.581	1.000e-6
MAPT-AS1 ★ ●	100128977	6.639	1.000e-6	8.312	1.000e-6	-0.994	0.004	2.270	0.002
NPFFR1	64106	-1.367	1.000e-6	2.046	0.009	1.294	1.434e-5	1.934	5.402e-6
OLAH	55301	-0.757	0.002	1.511	0.011	3.010	1.000e-6	3.751	1.000e-6
SRSF12	135295	-1.373	1.000e-6	0.758	0.034	0.908	1.000e-6	1.228	1.000e-6
TDRD1 ★	56165	2.053	1.533e-6	-1.427	0.015	0.961	7.994e-5	1.130	0.002
TPRXL	348825	-1.131	1.000e-6	0.887	0.003	0.967	0.002	0.923	6.105e-4
ZFP57	346171	-0.943	0.043	3.242	0.011	3.182	1.000e-6	3.974	1.000e-6
ZIC1 ●	7545	-2.364	1.000e-6	3.340	8.559e-5	1.869	2.257e-6	2.589	1.000e-6
ZP1	22917	0.853	0.021	-1.308	0.036	0.721	0.004	0.689	0.035

Figure 5. External validation and identification of domain-specific biomarkers. Differential expression (DE) analysis conducted via iPathwayGuide on the independent *TCGA BRCA* dataset provides an external benchmark for PAM50 molecular subtypes. The table lists significant subtype-specific genes that exhibit exclusive up- or down-regulation. To validate the harmonization process, we cross-referenced these external biomarkers with the findings from our adapted models: **Green circles** indicate genes identified in our **Source** (*SMC*) dataset that are validated by the *TCGA BRCA* benchmark. **Red stars** denote biomarkers identified in the **Target** (*CPTAC*) dataset following KPCA-DA adaptation that correspond to the external *TCGA BRCA* reference. The limited overlap between source and target biomarkers—while both remain consistent with the external *TCGA* benchmark—highlights the framework's ability to preserve the *intrinsic biological structure* of the target domain. Rather than erasing domain-specific variation through rigid alignment , KPCA-DA facilitates the discovery of unique biological drivers present in the target cohort that are independently verified by the external *TCGA* "gold standard".

CAN (58.0%)—suggests that study-specific artifacts in transcriptomics are inherently nonlinear. By projecting data into a high-dimensional Reproducing Kernel Hilbert Space (RKHS), KPCA-DA captures complex gene-gene interactions that linear models obscure.

Furthermore, a significant issue in unsupervised domain adaptation is that the model risks alignment achievement by squashing the target distribution, thereby losing biological heterogeneity. Our implementation of a Maximum-Variance Loss alongside MMD alignment ensures that while the study-specific domain effects are removed, the intrinsic biological variance of the tumor is preserved. This is evidenced by our success in the **TCGA-BRCA** validation tasks, where maintaining internal variance was critical for separating patient groups with distinct survival trajectories.

Our ablation analysis (Table 2) provides critical empirical confirmation regarding the mechanics of the Batch Effect Paradox. By validating that the combination of kPCA, MMD, and VML strictly outperforms any isolated sub-configuration, these results prove that our framework does not merely force an ar-

Table 4. Clinically significant biomarkers recovered via KPCA-DA. High-priority genes identified by Grad-CAM attributions in the target domain, validated against independent benchmarks.

Validation Benchmark	Key Recovered Genes (Grad-CAM)
PAM50 Signature	*BAG1*, *EGFR*, *FOXC1*, *EXO1*, *KRT14*, *SFRP1*, *ACTR3B*, *CEP55*
TCGA-DE (Target Discovery)	*BLACAT1*, *HS6ST3*, *TDRD1*, *MAPT-AS1**
Biological Insight	**Significance**
Phenotypic Accuracy	Recovery of ***EGFR/FOXC1*** validates Basal-like detection[23].
Mitotic Clock	Identification of *EXO1/CEP55* captures cell-cycle signals[24,25].
Target-Specific Discovery	Discovery of ***TDRD1, BLACAT1, HS6ST3*** unmasks markers of metastasis and genomic instability obscured prior to alignment[26,27].

*Conserved across both source and target domains.

tificial alignment between heterogeneous datasets. Instead, it safely unmasks a shared latent biological truth while maintaining site-specific feature hierarchies robustly across divergent experimental conditions.

Restoring the "White-Box" in Deep Learning

A central critique of deep domain adaptation frameworks like CAN[20] is their reliance on abstract latent spaces. While CAN achieved respectable statistical alignment, it effectively discards the explicit mapping between input genes and learned features, rendering the model a "black-box." In a clinical context, a prediction without a biological rationale is often unactionable for a physician.

Our framework's spatial preservation—embedding aligned features into structured *DeepInsight* image representations—enables a post-hoc "White-Box" audit via Grad-CAM. The recovery of clinical markers like *EGFR* and *FOXC1*[23] is not merely a technical achievement, reflecting that it is a validation of biological integrity. It proves that the adaptation process unmasked the latent biological truth previously hidden by study-specific noise.

Feature Restoration and Target-Specific Discovery

A primary contribution of this work is the demonstration of the model's capacity for target-specific feature recovery. While *MAPT-AS1* was identified as a conserved regulatory backbone across both the SMC (source) and CPTAC (target) domains, KPCA-DA enabled the identification of TCGA-validated markers in the target domain that were not prioritized in the source.

The identification of *BLACAT1*, *HS6ST3*, and *TDRD1*—established TCGA-validated biomarkers—exclusively within the adapted target data suggests that KPCA-DA does not merely impose the source's feature hierarchy onto the target domain. Instead, the framework restores the target dataset's signal-to-noise ratio to a degree that allows the model to capture valid, site-specific biological drivers.

Specifically, the recovery of the lncRNA *BLACAT1* and the cancer-testis antigen *TDRD1* indicates that KPCA-DA unmasks essential genomic markers inherent to the target cohort that were previously obscured by nonlinear domain shifts. *BLACAT1* has been identified as a critical epigenetic regulator that promotes proliferation and metastasis in breast cancer via the miR-150-5p/CCR2 axis[26], while *TDRD1* functions as a canonical cancer-testis antigen whose re-expression is a hallmark of genomic instability and aggressive phenotypes[27]. Furthermore, the identification of *HS6ST3* highlights the framework's ability to recover markers of the tumor microenvironment. As a heparan sulfate 6-O-sulfotransferase, its activity is critical for modulating growth factor signaling within the extracellular matrix[28]. The fact that these high-priority features were uniquely identified in the adapted target domain—despite not being primary drivers in the source—demonstrates that KPCA-DA restores the specific biological context of the target dataset. This enables the model to evolve from a generic classifier into a site-specific discovery tool. Ultimately, our framework anchors the data alignment to "Ground Truth" biology instead of relying on simple statistical matching.

Implications for Computational Oncology

The ability to perform alignment without the restrictive requirement for identical feature sets represents a significant step toward **Decentralized Decision Intelligence**. Most clinical trials are limited by small cohort sizes; the ability to seamlessly integrate a local cohort with global datasets like TCGA or CPTAC while maintaining interpretability allows for the development of more robust, scalable diagnostic tools.

While KPCA-DA demonstrates superior performance in transcriptomics, its reliance on a pre-defined kernel suggests that future iterations could benefit from Multiple Kernel Learning (MKL) to automatically learn optimal feature spaces for multi-omics integration. By providing a robust, unsupervised framework that harmonizes these nonlinear shifts, KPCA-DA enhances dataset interoperability and paves the way for reproducible, interpretable AI in clinical oncology.

Methodology

To effectively integrate gene expression datasets from multiple sources, it is crucial to employ methods that can learn to quantify and subsequently reduce the inherent distributional differences or "domain shifts" among them. For gene expression data, a fundamental prerequisite for this process is to establish a robust and consistent representation that enables the quantification of transcriptomic abundance. To address this, we propose KPCA-DA (Kernel Principal Component Analysis Based Domain Adaptation), an unsupervised method to align same-modality gene expression datasets while preserv-

ing biologically meaningful variation. kPCA is closely tied to the concept of Reproducing Kernel Hilbert Space (RKHS), as it operates implicitly within the RKHS defined by a chosen kernel function. By applying kPCA to both the source and target datasets, we facilitate the computation of Maximum Mean Discrepancy (MMD), which can subsequently be used to address and adapt the distributional differences between the two domains.

The KPCA-DA framework comprises three principal components, each addressing a specific facet of the data integration challenge:

Feature transformation via kPCA – Mapping gene expression data into a high-dimensional feature space to capture nonlinear relationships between genes. This is followed by the construction of a reduced-dimensionality representation.

Domain adaptation using Maximum Mean Discrepancy (MMD) – Quantifying and reducing distributional differences between datasets, ensuring alignment while retaining domain-specific information.

Image-based transformation using DeepInsight – By converting aligned datasets into a structured format suitable for deep learning, enabling CNN-based feature extraction and analysis.

The proposed multi-staged KPCA-DA pipeline synergistically combines kernel-based feature transformation, statistical domain alignment, and deep learning-compatible data restructuring facilitating the harmonization of gene expression datasets, improving model generalization across studies. The following sections provide a detailed explanation of each step in the KPCA-DA framework and its application to real-world genomic data.

Application of kPCA for Mapping Data to RKHS Space

Gene expression datasets are inherently high-dimensional, with complex and nonlinear interactions among genes that form intricate biological networks and pathways. These characteristics, coupled with domain shifts that arise when integrating data from multiple sources, necessitate the use of advanced analytical approaches. By employing a kernel function, kPCA implicitly maps the gene expression data into a new, higher-dimensional feature space where these complex, nonlinear dependencies can be more efficiently captured and analyzed. Standard Principal Component Analysis (PCA) finds a linear transformation that maximizes variance in the original feature space. However, traditional PCA often fails to properly capture the complex *nonlinear relationships* exhibited in biological data such as gene expression data. Kernel PCA (kPCA) extends PCA by mapping data into a **Reproducing Kernel Hilbert Space (RKHS)** via a nonlinear function $\phi(\cdot)$, allowing us to perform PCA in this transformed space[29].

RKHS is a vector space where functions can be represented as inner products using kernel functions, enabling us to model complex, nonlinear relationships between genes without explicitly computing their high-dimensional mappings. kPCA operates within RKHS, allowing us to transform gene expression data into a space where feature distributions can be better aligned across datasets. Instead of explicitly performing the nonlinear transformation and then computing the dot product, kPCA uses the **kernel trick** to compute similarities between genes efficiently in this space. We adopt kPCA over learned nonlinear alternatives because kPCA provides a deterministic, closed-form projection into RKHS via eigen-decomposition, capturing nonlinear gene interactions without the training stochasticity or initialization sensitivity of iteratively learned representations—an advantage in the small-sample, high-dimensional transcriptomic regime[30]. By applying kPCA to both the source and target datasets, we facilitate the computation of Maximum Mean Discrepancy (MMD), which can subsequently be used to address and adapt the distributional differences between the two domains.

We define the **source domain mRNA dataset** as:

$$X_s = \{S_s^m \times G_s^p\} \tag{1}$$

Here S_s represents the m samples and $G_s = \{g_1, g_2, ..., g_p\}$ is the p-dimensional gene feature set. Similarly, the **target domain dataset** is represented by:

$$X_t = \left\{S_t^n \times G_t^q\right\} \tag{2}$$

with S_t as the n samples and $G_t = \{g_1, g_2, ..., g_q\}$ as the q-dimensional gene feature set.

Unlike traditional kPCA, which applies dimensionality reduction to samples and maps them to the reduced space, we **transpose the dataset** and apply kPCA along the genes and map them on the reduced space. In this study, we are applying feature alignment based domain adaptation which focuses on aligning the feature distributions between the source and target domains, making the features appear similar across domains. The key idea is to "adapt" the feature space so that the models for downstream tasks see similar features regardless of the domain. The strength of feature alignment is that it allows the similar learned representations to apply across all domains[31]. In KPCA-DA, the genes G_s and G_t from the source and target datasets are mapped to a space where the respective genes from both sets can be represented similarly. First G_s and G_t are mapped to a higher-dimensional RKHS space using a nonlinear mapping function $\phi(g)$:

$$\Phi : \mathbb{R}^d \to \mathbb{R}^D, \quad \phi : \mathscr{X} \to \mathscr{H}, \quad g \mapsto \phi(g) \tag{3}$$

Here, $\mathbb{R}^d$ is the original feature space, and $\mathbb{R}^D$ is the high-dimensional RKHS space.

The **kernel trick** avoids the computational intractability of explicit computing of $\phi(g)$ by leveraging the implicit dot product representation in the transformed feature space. Gene-gene correlations g_i, g_j are computed through a kernel function K separately for source and target:

$$K_{ij}^s = K(g_i^s, g_j^s) = \langle \phi(g_i^s), \phi(g_j^s) \rangle_{\mathscr{H}} \tag{4}$$

$$K_{ij}^t = K(g_i^t, g_j^t) = \langle \phi(g_i^t), \phi(g_j^t) \rangle_{\mathscr{H}} \tag{5}$$

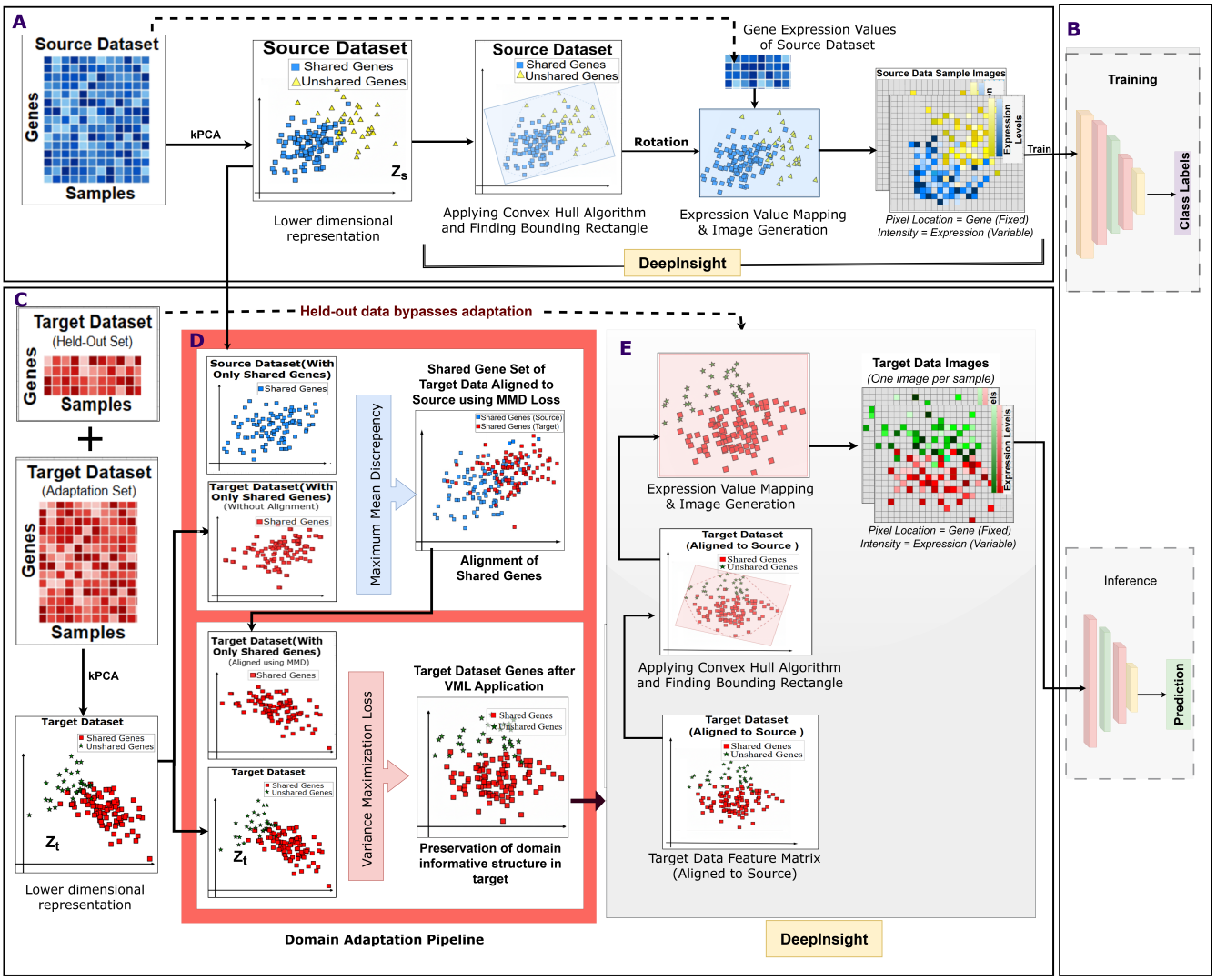


Figure 6. Schematic architecture of the KPCA-DA framework and DeepInsight-based sample encoding. The pipeline facilitates unsupervised domain adaptation by bridging kernel-based feature alignment with spatial deep learning. **(A, B) Source Transformation and Training:** Gene expression profiles are projected into a latent Reproducing Kernel Hilbert Space (RKHS) via **kPCA** to capture nonlinear interactions. The **DeepInsight** algorithm maps these 1D features into a fixed 2D grid, generating image-like representations where pixel intensity reflects expression levels for CNN training. **(C, D) Target Adaptation Mechanism:** The target dataset is split into an adaptation set (70%) and a held-out target-test set (30%). Optimization is conducted across two objectives: minimization of **Maximum Mean Discrepancy (MMD)** to align shared genes (red to blue) and maximization of **Variance Loss (VML)** to preserve the intrinsic structural relationship of unshared genes (green), preventing representation collapse. **(E) Inference Pipeline:** The held-out test data bypasses the adaptation training to ensure a rigorous generalization test. These samples are directly transformed into the learned image format for classification using the adapted ResNet-50 model.

where K is a pre-defined kernel function such as **Gaussian (RBF), polynomial, or sigmoid kernels**. The computed **kernel matrix** K encapsulates nonlinear gene interactions, making it ideal for domain adaptation.

To establish the **low-dimensional embedding**, the procedure requires the eigen decomposition of the centered kernel matrices K'_s and K'_t. Solving the corresponding eigenvalue problems yields the principal components that define the new axes of the feature space.

$$K'_s v_s = A_s v_s, \quad K'_t v_t = A_t v_t \tag{6}$$

where A represents the eigenvalues and v represents the eigenvectors. The **top-k eigenvectors** corresponding to the largest eigenvalues are selected to form the new reduced-dimensional representation:

$$Z_s = (X_s.T)^{\text{reduced}} = (V_s)_k (\Lambda_s)_k^{-\frac{1}{2}} \tag{7}$$

$$Z_t = (X_t.T)^{\text{reduced}} = (V_t)_k(\Lambda_t)_k^{-\frac{1}{2}} \tag{8}$$

where V_s and V_t are the selected eigenvectors, and Λ_s, Λ_t are the diagonal matrices containing the corresponding eigenvalues.

By applying kPCA, we obtain a transformed representation of **gene features in an RKHS space**, ensuring that *nonlinear relationships* among genes are preserved. The resultant nonlinear feature manifold will then serve as the input for the **domain adaptation step**, where we use **MMD-based optimization** to *minimize distributional differences between the empirical feature spaces of the source and target datasets.*.

Domain Adaptation with MMD and Variation Maximization

Our primary objective is to enforce statistical convergence between the feature distributions of the target and source datasets. Given the complex, nonlinear structure of gene expression data, minimizing the statistical difference between the target projections Z_t and the source projections Z_s in the principal component space can be effective. kPCA enables the projection of datasets into RKHS, where the MMD can be computed to measure domain discrepancy[13]. The adaptation is achieved through an iterative optimization process that dynamically modifies the target data's coordinates within the low-dimensional principal component subspace until the distributional discrepancy quantified by the MMD is rigorously minimized.

Measuring Distributional Discrepancy with MMD

The MMD framework provides a robust statistical test to determine if two samples are drawn from the same distribution. It operates by embedding the distributions into an infinite-dimensional RKHS, where the distance between their mean embeddings can be computed. MMD is a non-parametric statistical method suitable for high-dimensional data where traditional parametric assumptions are often invalid. It can quantify the divergence across all moments of the distribution, thereby capturing subtle, nonlinear shifts in the feature space[13]. Given two probability distributions P and Q, the MMD is defined as:

$$\text{MMD}^2(P, Q; \mathscr{H}) = \|\mu_P - \mu_Q\|^2_{\mathscr{H}} \tag{9}$$

where μ_P and μ_Q are the *mean embeddings* of distributions P and Q in the RKHS $\mathscr{H}$. The MMD term is squared here which defines the squared distance of the datasets in interest within the RKHS space. The squared MMD is typically smooth and differentiable making it more effective in optimization.

Similarly, the distributional discrepancy between the transformed source features Z_s and target features Z_t is formally quantified via the MMD, which serves as the adaptation loss function:

$$\mathscr{L} = \text{MMD}^2(Z_s, Z_t) \tag{10}$$

The squared MMD is the squared distance between the mean embeddings of the source and target distributions μ_{Z_s} and μ_{Z_t}, within the RKHS $\mathscr{H}$:

$$MMD^2(Z_s, Z_t) = ||\mu_{Z_s} - \mu_{Z_t}|| \tag{11}$$

Expanding the squared norm as an inner product gives-

$$\begin{aligned} ||\mu_{Z_s} - \mu_{Z_t}||^2_{\mathscr{H}} = \langle \mu_{Z_s} - \mu_{Z_t}, \mu_{Z_s} - \mu_{Zt} \rangle_{\mathscr{H}} = \\ \langle \mu_{Z_s}, \mu_{Z_s} \rangle_{\mathscr{H}} - 2\langle \mu_{Z_s}, \mu_{Z_t} \rangle_{\mathscr{H}} + \langle \mu_{Z_t}, \mu_{Z_t} \rangle_{\mathscr{H}} \end{aligned} \tag{12}$$

Here, the empirical estimates of these mean embeddings are given by $\hat{\mu}_{Z_s} = \frac{1}{p}\sum_{i=1}^{p} \phi(Z_{s_i})$ and $\hat{\mu}_{Z_t} = \frac{1}{q}\sum_{j=1}^{q} \phi(Z_{t_j})$. Substituting these empirical estimates into the expanded formula and leveraging the bi-linearity of the inner product yields:

$$\begin{aligned} MMD^2(Z_s, Z_t) = &\left\langle \frac{1}{p}\sum_{i=1}^{p} \phi(Z_{s_i}), \frac{1}{p}\sum_{j=1}^{p} \phi(Z_{s_j}) \right\rangle_{\mathscr{H}} \\ &-2\left\langle \frac{1}{p}\sum_{i=1}^{p} \phi(Z_{s_i}), \frac{1}{q}\sum_{j=1}^{q} \phi(Z_{t_j}) \right\rangle_{\mathscr{H}} \\ &+\left\langle \frac{1}{q}\sum_{i=1}^{q} \phi(Z_{t_i}), \frac{1}{q}\sum_{j=1}^{q} \phi(Z_{t_j}) \right\rangle_{\mathscr{H}} \end{aligned} \tag{13}$$

This simplifies to:

$$\begin{aligned} MMD^2(Z_s, Z_t) = &\frac{1}{p^2}\sum_{i=1}^{p}\sum_{j=1}^{p} \langle \phi(Z_{s_i}), \phi(Z_{s_j}) \rangle_{\mathscr{H}} \\ &-\frac{2}{pq}\sum_{i=1}^{p}\sum_{j=1}^{q} \langle \phi(Z_{s_i}), \phi(Z_{t_j}) \rangle_{\mathscr{H}} \\ &+\frac{1}{q^2}\sum_{i=1}^{q}\sum_{j=1}^{q} \langle \phi(Z_{t_i}), \phi(Z_{t_j}) \rangle_{\mathscr{H}} \end{aligned} \tag{14}$$

The "kernel trick" is then applied, which utilizes the reproducing property of the RKHS, where the inner product of the feature mappings of two points is equal to the evaluation of the kernel function on those points $\langle \phi(x), \phi(y) \rangle_{\mathscr{H}} = k(x, y)$. By replacing the inner products with the kernel function $k(\cdot, \cdot)$, we arrive at the final empirical estimate.

$$\begin{aligned} \text{MMD}^2(Z_s, Z_t) = &\frac{1}{p^2}\sum_{i,j} k(Z_{s_i}, Z_{s_j}) + \frac{1}{q^2}\sum_{i,j} k(Z_{t_i}, Z_{t_j}) \\ &-\frac{2}{pq}\sum_{i,j} k(Z_{s_i}, Z_{t_j}), \end{aligned} \tag{15}$$

Regularization for Variance Preservation

The source and target datasets often have differing feature spaces. For instance, the source dataset may contain 25,000 genes, while the target dataset has only 12,000. After independently applying kPCA to both datasets, the transformed feature matrices Z_s and Z_t may differ in dimensionality, making direct alignment infeasible.

One solution is to compute MMD only on the shared feature set across both datasets:

$$\mathscr{L} = \text{MMD}^2(Z_{s,\text{shared}}, Z_{t,\text{shared}}) \tag{16}$$

However, solely minimizing MMD in this manner risks *collapsing* the target data toward the source representation, potentially erasing biologically meaningful variations in the target domain.

To preserve the target dataset's structure, we introduce a **variance maximization regularization term**, which encourages the target distribution to retain its intrinsic variability while still aligning with the source distribution[32]. With the variance maximization loss (VML), the updated loss function is:

$$\mathscr{L} = \mathrm{MMD}^2(Z_{s,\mathrm{shared}}, Z_{t,\mathrm{shared}}) + \lambda \cdot \mathrm{trace}(Z_t^\top K_t Z_t) \quad (17)$$

where the trace term ensures that unshared features in Z_t retain their internal structure by maintaining consistency with the target kernel K_t.

The optimization objective is to minimize $\mathscr{L}$, thereby aligning the target projections with the source domain while preserving the target data structure:

$$\min \mathscr{L} = \mathrm{MMD}^2(Z_{s,\mathrm{shared}}, Z_{t,\mathrm{shared}}) + \lambda \cdot \mathrm{trace}(Z_t^\top K_t Z_t) \quad (18)$$

The first term ensures that shared features between Z_s and Z_t are aligned, while the second term preserves structural information in Z_t.

Optimization of The Adaptation Process

The minimization of the composite loss function, $\mathscr{L}$, is achieved through a gradient-based optimization scheme. Since both terms have different optimization objectives and do not share common parameters, we compute their gradients separately, and the overall gradient is then formulated as the linear combination of these individual gradients, which guides the iterative update of the target feature space.

The goal of the optimization is to modify the target feature representation $Z_{t,shared}$, so that its distribution moves closer to the source distribution $Z_{s,shared}$. The gradient of the MMD term with respect to $Z_{t,\mathrm{shared}}$ is given by:

$$\begin{aligned}\nabla_{Z_{s,\mathrm{shared}}} \mathrm{MMD}^2 = &\frac{1}{n_s^2}\sum_{i,j}\frac{\partial}{\partial Z_{s,\mathrm{shared}}}k(Z_{s,\mathrm{shared}_i}, Z_{s,\mathrm{shared}_j}) \\ &+\frac{1}{n_t^2}\sum_{i,j}\frac{\partial}{\partial Z_{t,\mathrm{shared}}}k(Z_{t,\mathrm{shared}_i}, Z_{t,\mathrm{shared}_j}) \\ &-\frac{1}{n_s n_t}\sum_{i,j}\frac{\partial}{\partial Z_{t,\mathrm{shared}}}k(Z_{s,\mathrm{shared}_i}, Z_{t,\mathrm{shared}_j})\end{aligned} \quad (19)$$

The source feature representation $Z_{s,shared}$ here is considered fixed and does not change during this process. However, as Z_t moves closer to Z_s during optimization, the overall MMD decreases due to the cross domain kernel term between Z_t and Z_s. This gradient encapsulates the two primary forces of adaptation: the second term models the interactions within the target domain, while the third term models the interactions between the source and target domains, acting as the primary alignment force that pulls the target distribution toward the source.

The gradient for the variance maximization term with respect to Z_t is defined as:

$$\nabla_{Z_t} \lambda \cdot \mathrm{trace}(Z_t^\top K_t Z_t) = 2\lambda K_t Z_t \quad (20)$$

Thus, the final gradient for the adaptation loss $\mathscr{L}$ is:

$$\nabla_{Z_t}\mathscr{L} = \nabla_{Z_t}\mathrm{MMD}^2(Z_s, Z_t) + \lambda \cdot \nabla_{Z_t}\mathrm{trace}(Z_t^\top K_t Z_t) \quad (21)$$

The update for Z_t at each iteration c is performed using gradient descent:

$$Z_t^{(k+1)} = Z_t^{(k)} - \eta_k \nabla_{Z_t}\mathscr{L}(Z_s, Z_t) \quad (22)$$

By iteratively applying this update, we achieve an optimal alignment of target projections with the source projections while maintaining essential structure in the target dataset. This ensures that the domain-adapted target dataset retains its biological relevance while improving generalizability across different gene expression datasets. Following the alignment, the source and the domain-adapted target datasets are then processed by the DeepInsight module, converting them into a two-dimensional image format for subsequent analysis.

Sample-Level Image Conversion using DeepInsight

DeepInsight is a technique designed to convert structured non-image data, such as gene expression feature vectors, into a grid-like format suitable for deep learning models, particularly Convolutional Neural Networks (CNNs)[15]. Unlike traditional feature-based models, CNNs leverage spatial relationships in data, making the arrangement of features within the transformed grid crucial for accurate pattern recognition.

Since the gene features have already been aligned in the previous kPCA-based domain adaptation step, the spatial arrangement remains unchanged across samples. DeepInsight ensures that only expression intensity varies while maintaining the biologically relevant structure of the features. Each sample is transformed into a single image, where genes occupy the same spatial positions across all samples, and pixel intensity levels reflect expression values specific to each sample.

The transformation process begins by positioning genes on a 2D grid based on pairwise similarity, ensuring biologically related genes remain spatially close. A fixed spatial layout is maintained across all samples, allowing CNNs to focus on variations in expression intensity rather than changes in gene placement. Each gene's expression level is then mapped to an intensity value, forming an image-like representation where differences in pixel intensity correspond to differences in gene expression levels. The resulting images are then used as input for CNN-based models, such as ResNet, which extract high-level spatial feature patterns and leverage deep learning techniques for subsequent classification tasks.

Experimental Setup

To evaluate the efficacy of **KPCA-DA**, we implemented a rigorous cross-domain evaluation protocol designed to simulate real-world generalization. For each task, the target domain was partitioned into a *target-train* subset (70%) and a held-out *target-test* subset (30%). Critically, the target-test set was excluded from all alignment and training procedures, serving exclusively for final performance evaluation. This stricter setting ensures that the observed improvements reflect the model's

ability to generalize to unseen target samples encountered post-deployment. Details of the preprocessing pipeline, including variance filtering, feature scaling, and probe-to-gene symbol conversion for the Desmedt microarray dataset are provided in Supplementary Material, Note 3.

The KPCA-DA pipeline proceeds in two distinct phases: feature alignment and image-based classification (Fig. 6). Upon completion of the alignment (Panels C-D), a harmonized feature matrix is obtained for both domains (Panel E). We then employ *DeepInsight*[15] to transform these transcriptomic profiles into structured 2D image representations, where **each resulting image represents a single biological sample**. In this mapping, the spatial arrangement (pixel coordinates) identifies specific genes based on their similarity in the feature space, while the pixel intensity encodes the corresponding gene expression levels. These images serve as inputs for a **ResNet50** convolutional neural network. Specifically, the source domain images are used to train the classifier, which is subsequently evaluated on the images generated from the 30% held-out target-test set. Kernel bandwidth γ and regularization weight λ were both selected using the source dataset and the target adaptation subset, prior to evaluation on the held-out target-test set, via the procedures detailed in *supplementary material I*'s note 1 and note 2 respectively. It is important to note that because the spatial locations (pixel coordinates) for each gene are fixed during the initial DeepInsight mapping, generating an image for a new testing sample only requires mapping its specific transcriptomic intensities onto the established grid. This consistency ensures that the feature hierarchy learned by the ResNet50 model remains spatially aligned across both training and deployment phases.

To ensure a controlled comparison, we applied identical image-conversion and classification pipelines to the competing methods:

- **ComBat:** Batch effect parameters were estimated using the source and target-train datasets and subsequently applied to the target-test set to ensure consistent alignment prior to DeepInsight conversion.
- **CAN:** Unlike the feature-space alignment used in KPCA-DA and ComBat, CAN performs contrastive alignment in the image domain. Source and target datasets were first converted to images, the CAN feature extractor was then trained on the target-train set to learn domain-invariant representations, which were finally frozen and applied to the target-test set.
- **Baseline:** DeepInsight was applied directly to the raw source and target-test data without any intervening alignment or correction, providing a lower-bound for cross-domain performance.

This experimental design allows for a direct comparison between feature-space adaptation (KPCA-DA, ComBat) and latent image-domain adaptation (CAN), while highlighting the necessity of alignment via the unadapted baseline.

Conclusion

In this work, we introduced **KPCA-DA**, a nonlinear domain adaptation framework designed to overcome the persistent challenge of distributional shift in multi-source transcriptomic integration. By leveraging the high-dimensional feature mapping of the Reproducing Kernel Hilbert Space (RKHS) alongside a maximum-variance loss, our method achieves a significant reduction in distributional discrepancy while preserving the underlying biological manifold.

Unlike black-box deep learning approaches, the spatial preservation inherent in KPCA-DA enables post-hoc interpretability via Grad-CAM. This "White-Box" capability allowed for the restoration of canonical biomarkers like *EGFR* and the discovery of target-specific drivers such as *BLACAT1* and *TDRD1*. These findings confirm that our framework aligns datasets by unmasking shared biological truths rather than fitting to study-specific noise.

It is important to note that the current evaluation is conducted exclusively on breast cancer transcriptomic datasets, and the generalizability of KPCA-DA to other cancer types, disease contexts, or broader omics modalities (e.g., proteomics or DNA methylation) remains to be established in future work. In addition, all benchmarks in this study are derived from mRNA expression data, and performance characteristics may vary when the framework is applied to data with fundamentally different distributional properties.

Within these constraints, however, KPCA-DA provides a promising and interpretable unsupervised framework for multi-source genomic data harmonization. By enabling the integration of heterogeneous transcriptomic cohorts while preserving biological interpretability, the proposed approach represents a meaningful step toward more robust and potentially clinically relevant analytical tools in precision medicine, contingent upon further validation across diverse disease settings.

Author Contributions Statement

Shelia Rahman Tuly: Methodology, Investigation, Formal analysis, Writing – original draft. **Seeya Awadhut Munj:** Data analysis, Writing – review and editing. **Melike Yildirim:** Investigation, Writing – review and editing. **Alper Murat:** Investigation, Writing – review and editing. **Sorin Draghici:** Investigation, Writing – review and editing. **Suzan Arslanturk:** Conceptualization, Methodology, Investigation, Data curation, Supervision, Funding acquisition, Writing – original draft.

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Code Availability

The source code will be publicly available at GitHub upon acceptance.

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